

Matteo Leonelli

Cybersecurity Researcher

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Summary

Ph.D. candidate at CISA specializing in coverage-guided fuzzing, vulnerability research, and binary analysis, with peer-reviewed publications at top security venues (NDSS, EuroS&P) and industry research experience at Amazon Web Services (AWS). Discovered and disclosed 4+ CVEs across hardware (Intel graphics), multimedia (FFmpeg), and widely-deployed WordPress plugins. Organizing committee member for the SBFT tool competition (ICSE 2025) and mentor to Bachelor's and undergraduate research students.

Experience

CISPA Helmholtz Center for Information Security, Saarland University *Saarbrücken, Germany*
PH.D. CANDIDATE, SYSTEM SECURITY *Mar. 2024 – Present*

- EuroS&P'26: **THEMIS** – context-aware grey-box fuzzer for WordPress plugins; discovered multiple authenticated/unauthenticated vulnerabilities in widely-deployed plugins
- NDSS'25: **TwinFuzz** – differential testing framework for video hardware acceleration stacks, leading to Intel CVE-2024-23919
- SBFT'25: co-organized the Fuzzing Track Tool Competition at ICSE 2025
- Discovered and disclosed 4 CVEs across Intel graphics software and WordPress plugins (CVE-2024-23919, CVE-2025-9219, CVE-2025-5701, CVE-2023-7306)

RESEARCH ASSISTANT *Apr. 2022 – Mar. 2024*

- Collaborated with Ph.D. researchers on automated root-cause analysis of fuzzer-discovered bugs, localizing vulnerabilities to specific commits and code versions across software histories
- Researched LLM-driven automatic fuzzing-harness generation to reduce manual effort in setting up fuzzing targets across different frameworks

MASTER'S THESIS *Mar. 2023 – Jan. 2024*

- Research on software and hardware fuzzing in collaboration with Intel

Amazon Web Services (AWS) *Paris, France*
APPLIED SCIENTIST INTERN – RESEARCH INTERNSHIP *Sep. 2025 – Nov. 2025*

- Designed and implemented automated fuzz-testing pipelines to evaluate and ensure data integrity in Amazon S3
- Applied grey-box fuzzing techniques to cloud storage systems, identifying edge-case failure modes in large-scale distributed infrastructure

Fondazione Bruno Kessler *Trento, Italy*
BACHELOR THESIS *May 2021 – Sep. 2021*

- Research resulting in the CODASPY'21 paper: *Secure Pull Printing with QR Codes and National eID Cards: A Software-Oriented Design and an Open-Source Implementation*

JUNIOR RESEARCHER *Feb. 2019 – Dec. 2020*

- Designed authentication protocols using the Italian CIE 3.0 smart card standard

SOFTWARE DEVELOPER INTERN, SECURITY & TRUST LAB *Nov. 2018 – Jan. 2019*

- Mobile developer internship focused on secure application development

JAVA DEVELOPER INTERN, SMART COMMUNITY LAB *Jun. 2017 – Jul. 2017*

- Java developer involved in data analysis and software development

Bitahoy *Saarbrücken, Germany*
IOT SECURITY RISK ASSESSMENT – CYSEC PROJECT *May 2021 – Aug. 2021*

- Assessed IoT device security using Wireshark to flag malicious traffic and cluster suspicious network activity

Matrix Tech *Trento, Italy*
FULL STACK DEVELOPER *Dec. 2017 – Dec. 2020*

- Developed a full-stack web service for access control and real-time occupancy monitoring, integrated with proprietary embedded hardware

OGP *Trento, Italy*
DEVELOPER *2018 – 2019*

- Developed customer-facing websites and integrated new features

Education

CISPA Helmholtz Center for Information Security *Saarbrücken, Germany*
PH.D. IN CYBERSECURITY *Mar. 2024 – Present*

Saarland University *Saarbrücken, Germany*
M.ENG. IN CYBERSECURITY *Oct. 2021 – Feb. 2024*

University of Trento *Trento, Italy*
B.SC. IN COMPUTER SCIENCE *Sep. 2018 – Sep. 2021*

Technological Institute "G. Marconi" *Rovereto, Italy*
DIPLOMA IN COMPUTER SCIENCE AND TELECOMMUNICATIONS *Sep. 2013 – Jul. 2018*

Skills

Research Interests

Coverage-Guided Fuzzing, Differential Testing, Binary Analysis, Web Application Security, Hardware Security Testing, Embedded Systems, Distributed Systems

Programming

C, C++, Python, Bash, Rust, PHP

Languages

Italian (Native), English (C1), German (B1), Spanish (B1)

Publications

THEMIS: Context-Aware Grey-box Fuzzing for WordPress Plugins

Matteo Leonelli, David Jannis Dewes, Thorsten Holz

IEEE EuroS&P
2026

SBFT Tool Competition 2025 – Fuzzing Track

Addison Crump, Sahil Sihag, Matteo Leonelli

SBFT
2025

TwinFuzz: Differential Testing of Video Hardware Acceleration Stacks

Matteo Leonelli, Addison Crump, Meng Wang, Florian Bauckholt, Keno Hassler, Ali Abbasi, Thorsten Holz

NDSS
2025

Secure Pull Printing with QR Codes and National eID Cards: A Software-Oriented Design and an Open-Source Implementation

Matteo Leonelli, Umberto Morelli, Giada Sciarretta, Silvio Ranise

ACM CODASPY
2021

Pull Printing with National eID Cards: An Open-Source and Software-Oriented Implementation

Matteo Leonelli, Umberto Morelli, Silvio Ranise, Giada Sciarretta

ITASEC
2020

Vulnerabilities & Achievements

2025 CVE-2025-9219

Post SMTP (WordPress plugin) – missing authorization to authenticated subscriber

2025 CVE-2025-5701

HyperComments (WordPress plugin) – unauthenticated arbitrary options update

2025 7 more findings, WordPress plugins

Modern Events Calendar, rtMedia, URL Shortener Plugin – LFI, SQLi, SSRF, PHP object injection, option update

2024 CVE-2024-23919

Improper buffer restrictions in Intel Graphics software (Intel Security Acknowledgements)

2024 Confirmed, FFmpeg

Free() attempted on address that was never malloc'ed

2024 Confirmed, Firefox

Cross-tab info leak via hardware-accelerated decoding (confirmed by Mozilla)

2024 Reported, VLC

Crash via Direct3D11 interaction on hardware-accelerated decoding

2023 CVE-2023-7306

Frontend File Manager (WordPress plugin) – unauthenticated arbitrary post deletion

2023 Confirmed, FFmpeg

Global buffer overflow

Teaching & Supervision

2025–26 Research Assistant Mentor

CISPA Helmholtz Center for Information Security

2025 Bachelor's Thesis Supervisor

CISPA Helmholtz Center for Information Security

2024 Teaching Assistant, System Security

CISPA Helmholtz Center for Information Security

2023 Teaching Assistant, Security

University of Saarland

2023 Teaching Assistant, Human Computer Interaction

University of Saarland

2021 Teaching Assistant, Programming 1

University of Trento

Talks & Summer Schools

2026 Attended IEEE S&P San Francisco, USA

2026 Attended Fuzzing and Software Security Summer School Singapore

2026 Invited Contributor, CISPA Blog Article: Themis CISPA

2025 Attended and Presented at NDSS San Diego, USA

2025 Organized SBFT Workshop ICSE 2025

2025 Invited Guest, CISPA Podcast Black Box Testing with Matteo Leonelli

2024 Attended and Presented at Intel Workshop: Scalable Assurance Portland, USA

2024 Attended EPFL Summer School in System Security Lausanne, Switzerland

2023 Attended CISPA Summer School in System Security Saarland, Germany

2022 Attended CISPA Summer School in Trustworthy Artificial Intelligence Saarland, Germany

2021 Attended University of Trento School of Innovation Trento, Italy

2019 Organized ProM Camp Trento, Italy

2018 Attended ProM Camp Trento, Italy